

```
#!/bin/bash

db="max30102.db"
table="heart_rate_data"

echo "—— WELCOME ——"

while [ 1 ]
do
    echo "Please select any query by typing its sr. no."

    echo " "
    echo "1. View heart rate data (last 50 entries) of ... "
    echo "2. Calculate average heart rate of ... "
    echo "3. Compare average heart of users."
    echo "4. View oxygen saturation (SpO2) data (last 50 entries) of ... "
    echo "5. Calculate average SpO2 of ... "
    echo "6. Compare average SpO2 of all users."

    echo " "
    echo -n "Enter sr. no. of query: "
    read choice

    if [ "$choice" = "" ];then echo "Value needed"; exit 1; fi

    case $choice in
        1)
            echo " "
            echo "Select (complete the sentence):"
            echo "    1. ...a person (enter name)."
            echo "    2. ...all users."
            echo " "
            echo -n "Enter your choice: "
            read ans

            if [ "$ans" = 1 ]; then
                echo -n "Enter person name: "
                read person

                echo " "
                sqlite3 -header -column "$db" "SELECT timestamp,
bpm FROM $table WHERE person='$person' ORDER BY id DESC LIMIT 50;"

            else
                echo "Displaying heart rate for all users."
                echo " "
                sqlite3 -header -column "$db" "SELECT timestamp,
bpm FROM $table ORDER BY id DESC LIMIT 50;"
            fi
        *)
            echo "Invalid choice. Please try again."
        esac
    done
```

```
fi
;;

2)

echo " "
echo -n "Calculate average HR for: "
read person

echo " "
sqlite3 -header -column "$db" "SELECT AVG(bpm) FROM $table
WHERE person='$person';"
;;

3)

echo " "
echo "Comparing average heart rate of all users:"

echo " "
sqlite3 -header -column "$db" "SELECT person, AVG(bpm) FROM
$table GROUP BY person;"
;;

4)

echo " "
echo "Select (complete the sentence):"
echo "    1. ...a person (enter name)."
echo "    2. ...all users."
echo " "
echo -n "Enter your choice: "
read ans

if [ "$ans" = 1 ]; then
    echo -n "Enter person name: "
    read person

    echo " "
    sqlite3 -header -column "$db" "SELECT timestamp,
spo2 FROM $table WHERE person='$person' ORDER BY id DESC LIMIT 50;"

else
    echo "Displaying SpO2 data for all users."

    echo " "
    sqlite3 -header -column "$db" "SELECT timestamp,
person, spo2 FROM $table ORDER BY id DESC LIMIT 50;"

fi
;;
```

```
5)
    echo " "
    echo -n "Enter name of person: "
    read person

    echo " "
    sqlite3 -header -column "$db" "SELECT AVG(spo2) as
Average_SpO2 FROM $table WHERE person='$person';"
    ;;

6)
    echo " "
    echo "Comparing average SpO2 of all users:"

    echo " "
    sqlite3 -header -column "$db" "SELECT person, AVG(spo2) as
Average_SpO2 FROM $table GROUP BY person;"
    ;;

7)
    echo " "
    echo -n "Enter name of person: "
    read person

    echo " "
    sqlite3 -header -column "$db" "SELECT MAX(bpm) AS max_bpm,
MIN(bpm) AS min_bpm, AVG(bpm) AS avg_bpm, MAX(spo2) AS max_spo2, MIN(spo2) AS
min_spo2, avg(spo2) AS avg_spo2 FROM $table WHERE person='$person'"
    ;;

*)
    echo
    echo "Invalid selection. Please enter a number between 1
and 6."

    ;;

esac

echo " "
echo -n "Do you want to continue? (Y/n): "
read t_choice

if [ "$t_choice" = 'n' ]; then echo "Exiting program..."; exit 0;
else clear; fi;

done
```

```
rasbhari-pi remote shell - Raspberry Pi Connect - Personal - Microsoft Edge
https://connect.raspberrypi.com/devices/5b879d2b-0c76-426b-ab3e-8b8636789ccb/remote-shell-session
GNU nano 8.4 queries_script.sh
#!/bin/bash

db="max30102.db"
table="heart_rate_data"

echo "---- WELCOME ----"

while [ 1 ]
do
    echo "Please select any query by typing its sr. no."

    echo " "
    echo "1. View heart rate data (last 50 entries) of..."
    echo "2. Calculate average heart rate of..."
    echo "3. Compare average heart of users."
    echo "4. View oxygen saturation (SpO2) data (last 50 entries) of..."
    echo "5. Calculate average SpO2 of..."
    echo "6. Compare average SpO2 of all users."

    echo " "
    echo -n "Enter sr. no. of query: "
    read choice

    if [ "$choice" = "" ];then echo "Value needed"; exit 1; fi

    case $choice in
        1)

```

This code is written in Nano text editor whilst connected to Raspberry-Pi Model 2B via Raspberry-Pi Connect.

SSH was also possible but had to keep switching from terminal to screen (of RPi) because of Node-Red.

This is one of the queries being executed. The task is run by typing './<file_name>.sh' in the command line (Terminal).

It isn't always possible to run the script after writing it, first it has to be an executable. That was done by the command 'chmod +x <file_name>.sh' or 'chmod 777 <file_name>.sh'.

```
rasbhari-pi remote shell - Raspberry Pi Connect - Personal - Microsoft Edge
https://connect.raspberrypi.com/devices/5b879d2b-0c76-426b-ab3e-8b8636789ccb/remote-shell-session
pisylon@rasbhari-pi:/var/www/html/database $ ./queries_script.sh
---- WELCOME ----
Please select any query by typing its sr. no.

1. View heart rate data (last 50 entries) of...
2. Calculate average heart rate of...
3. Compare average heart of users.
4. View oxygen saturation (SpO2) data (last 50 entries) of...
5. Calculate average SpO2 of...
6. Compare average SpO2 of all users.

Enter sr. no. of query: 2

Calculate average HR for: Kaustubh

AVG(bpm)
-----
148.108108108108

Do you want to continue? (Y/n): n
Exiting program...
pisylon@rasbhari-pi:/var/www/html/database $
```

```
rasbhari-pi remote shell - Raspberry Pi Connect - Personal - Microsoft Edge
https://connect.raspberrypi.com/devices/5b879d2b-0c76-426b-ab3e-8b8636789ccb/remote-shell-session
pisylon@rasbhari-pi:/var/www/html/database $ ./queries_script.sh
---- WELCOME ----
Please select any query by typing its sr. no.

1. View heart rate data (last 50 entries) of...
2. Calculate average heart rate of...
3. Compare average heart of users.
4. View oxygen saturation (SpO2) data (last 50 entries) of...
5. Calculate average SpO2 of...
6. Compare average SpO2 of all users.

Enter sr. no. of query: 3

Comparing average heart rate of all users:

person    AVG(bpm)
-----
Kaustubh  148.108108108108

Do you want to continue? (Y/n):
```

More people can be added to this database. In-fact, the Node-Red utility is ready to implement more than 1 person, but there was shortage of hardware like ESP-8266 (due to other projects), so that couldn't be implemented in the given timeframe.